Represention of DELPH-IN Pragmatics

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Pragmatic Phenomena

There are several **discourse-related items** that can improve language processing systems.

- information structure
- honorification
- different types of questions: rhetoric questions, tag-questions, ...
- anaphora resolution within an intersentential domain
- ...

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Pragmatic Notions

- presupposition

(1) a. Kim has two children that study linguistics.

b. Kim has two children, who study linguistics.

- assertion, denial, question (SFORCE)
  - MA questions vs. A-NOT-A questions in Chinese
  - rhetoric questions: *Who knows?*
- speaker/hearer

“ The name is familiar to me , ” said he .

INNER FRAME

OUTER FRAME

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DELPH-IN Pragmatics

- http://moin.delph-in.net/JacyPragmatics
- unresolved issues
  - speaker/hearer scope in quotations
  - best way to handle polarity in honorification
  - topic
- The question is how to make pragmatic components **visible in semantic representation** with an eye toward semantics-based language processing

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MRS+ICONS

$$
\begin{bmatrix}
\textit{mrs+icons} & & \\
\text{HOOK} & \begin{bmatrix}
\textit{hook} & \\
\text{GTOP} & \textit{handle} \\
\text{LTOP} & \textit{handle} \\
\text{INDEX} & \textit{individual} \\
\text{XARG} & \textit{individual} \\
\text{ICONS-KEY} & \textit{icons} \\
\text{SPEAKER-KEY} & \textit{ref-ind} \\
\text{HEARER-KEY} & \textit{ref-ind}
\end{bmatrix} \\
\text{RELS} & \textit{diff-list} \\
\text{HCONS} & \textit{diff-list} \\
\text{ICONS} & \left\langle\, ! \ldots, \begin{bmatrix}
\textit{icons} & \\
\text{IARG1} & \textit{individual} \\
\text{IARG2} & \textit{individual}
\end{bmatrix}, \ldots\ ! \,\right\rangle
\end{bmatrix}
$$

ConTeXT

$$
\begin{bmatrix}
\textit{local} & \\
\text{CAT} & \textit{cat} \\
\text{CONT} & \textit{mrs} \\
\text{CTXT} & \begin{bmatrix}
\text{ACTIVATED} & \textit{bool} \\
\text{PRESUP} & \textit{diff-list} \\
\text{C-INDICES} & \begin{bmatrix}
\text{SPEAKER} & \textit{ref-ind} \\
\text{HEARER} & \textit{ref-ind}
\end{bmatrix}
\end{bmatrix}
\end{bmatrix}
$$

Utterance

$$\begin{bmatrix} \textit{utterance} & \\ \text{UTTERED} & + \\ \text{CONT} & \begin{bmatrix} \text{SPEAKER-KEY} & \boxed{1} \\ \text{HEARER-KEY} & \boxed{2} \end{bmatrix} \\ \text{CTXT} & \begin{bmatrix} \text{SPEAKER} & \boxed{1} \\ \text{HEARER} & \boxed{2} \end{bmatrix} \\ \text{ARGS} & \left\langle \mathbf{H} \begin{bmatrix} \textit{sat-or-frag} & \\ \text{UTTERED} & - \\ \text{INDEX} & \boxed{3} \end{bmatrix} \right\rangle \\ \text{ICONS} & \left\langle \begin{bmatrix} \textit{addressor} & \\ \text{IARG1} & \boxed{1} \\ \text{IARG2} & \boxed{3} \end{bmatrix}, \begin{bmatrix} \textit{addressee} & \\ \text{IARG1} & \boxed{2} \\ \text{IARG2} & \boxed{3} \end{bmatrix} \right\rangle \end{bmatrix}$$

(2) "You have been cruelly used," said Holmes.

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Sample Representation

(3)

お爺さん	が	お休み	に なり	ます
o-jiisan	ga	o-yasumi	ni nari	masu
HON-elderly.man	NOM	HON-sleep	become	HON

'The elderly man is sleeping.' (to an honoree) [jpn]

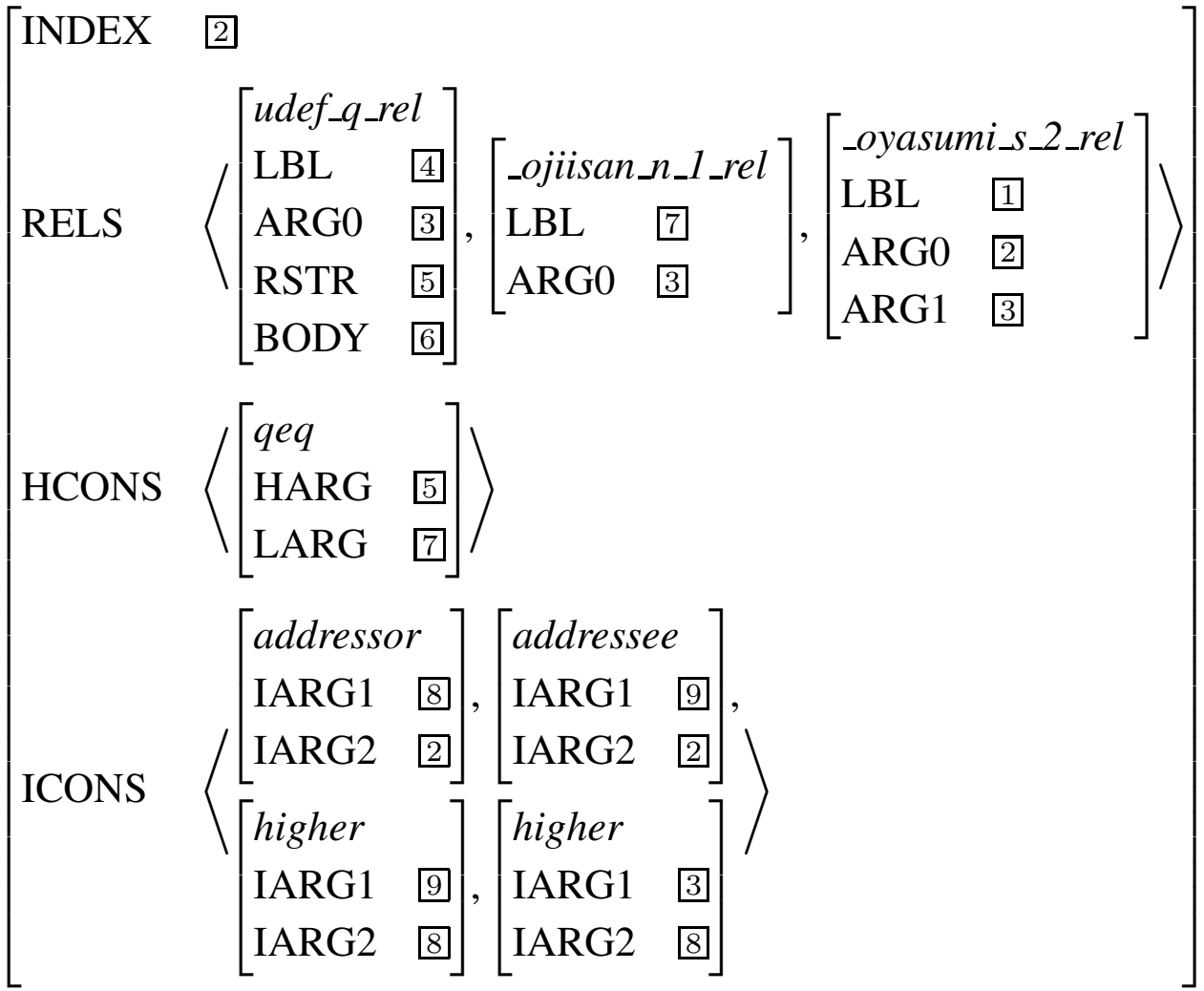
(4) a. *ojiisan* *oyasumi*
*x*1 *e*2

b. speaker: *x*3, hearer: *x*4

c. [*x*3 **addressor** *e*2] (utterance)
[*x*4 **addressee** *e*2] (utterance)
[*x*4 **higher** *x*3] (ます *masu*)
[*x*1 **higher** *x*3] (に なる *ni naru*)

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Sample Representation (cont'd)



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Sample Representation (cont'd)

